



Module 17: Transactions

Database System Concepts, 7th Ed.

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Outline

- Transaction Concept
- Transaction State
- Concurrent Executions
- Serializability
- Recoverability
- Implementation of Isolation
- Transaction Definition in SQL
- Testing for Serializability.



Transaction Concept

- A **transaction** is a *unit* of program execution that accesses and possibly updates various data items.
- E.g., transaction to transfer \$50 from account A to account B:
 1. **read**(A)
 2. $A := A - 50$
 3. **write**(A)
 4. **read**(B)
 5. $B := B + 50$
 6. **write**(B)
- Two main issues to deal with:
 - Failures of various kinds, such as hardware failures and system crashes
 - Concurrent execution of multiple transactions



Example of Fund Transfer

- Transaction to transfer \$50 from account A to account B:
 1. **read**(A)
 2. $A := A - 50$
 3. **write**(A)
 4. **read**(B)
 5. $B := B + 50$
 6. **write**(B)
- **Atomicity requirement**
 - If the transaction fails after step 3 and before step 6, money will be “lost” leading to an inconsistent database state
 - Failure could be due to software or hardware
 - The system should ensure that updates of a partially executed transaction are not reflected in the database
- **Durability requirement** — once the user has been notified that the transaction has completed (i.e., the transfer of the \$50 has taken place), the updates to the database by the transaction must persist even if there are software or hardware failures.



Example of Fund Transfer (Cont.)

- **Consistency requirement** in above example:
 - The sum of A and B is unchanged by the execution of the transaction
- In general, consistency requirements include
 - Explicitly specified integrity constraints such as primary keys and foreign keys
 - Implicit integrity constraints
 - e.g., sum of balances of all accounts, minus sum of loan amounts must equal value of cash-in-hand
 - A transaction must see a consistent database.
 - During transaction execution the database may be temporarily inconsistent.
 - When the transaction completes successfully the database must be consistent
 - Erroneous transaction logic can lead to inconsistency



Example of Fund Transfer (Cont.)

- **Isolation requirement** — if between steps 3 and 6, another transaction T2 is allowed to access the partially updated database, it will see an inconsistent database (the sum $A + B$ will be less than it should be).

T1	T2
1. read (A)	
2. $A := A - 50$	
3. write (A)	
	read(A), read(B), print(A+B)
4. read (B)	
5. $B := B + 50$	
6. write (B)	

- Isolation can be ensured trivially by running transactions **serially**
 - That is, one after the other.
- However, executing multiple transactions concurrently has significant benefits, as we will see later.



ACID Properties

A **transaction** is a unit of program execution that accesses and possibly updates various data items. To preserve the integrity of data the database system must ensure:

- **Atomicity.** Either all operations of the transaction are properly reflected in the database or none are.
- **Consistency.** Execution of a transaction in isolation preserves the consistency of the database.
- **Isolation.** Although multiple transactions may execute concurrently, each transaction must be unaware of other concurrently executing transactions. Intermediate transaction results must be hidden from other concurrently executed transactions.
 - That is, for every pair of transactions T_i and T_j , it appears to T_i that either T_j finished execution before T_i started, or T_j started execution after T_i finished.
- **Durability.** After a transaction completes successfully, the changes it has made to the database persist, even if there are system failures.



Storage Structure

- Volatile
- Non Volatile
- Stable

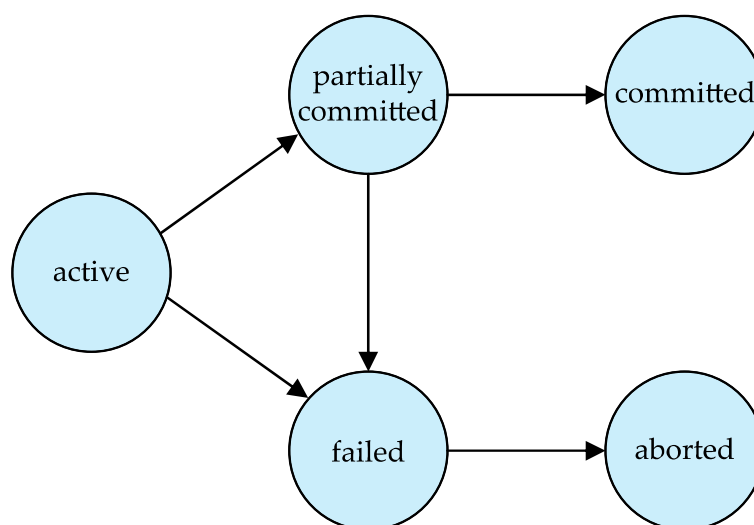


Transaction State

- **Active** – the initial state; the transaction stays in this state while it is executing
- **Partially committed** – after the final statement has been executed.
- **Failed** -- after the discovery that normal execution can no longer proceed.
- **Aborted** – after the transaction has been rolled back and the database restored to its state prior to the start of the transaction. Two options after it has been aborted:
 - Restart the transaction
 - Can be done only if no internal logical error
 - Kill the transaction
- **Committed** – after successful completion.



Transaction State (Cont.)





Additional Issues

- Rollback is achieved by log
- External writes
 - What should a restarted AM Machine do?
 - What should a ticket booking site do?
- Compensating Transaction



Concurrent Executions

- Multiple transactions are allowed to run concurrently in the system.
Advantages are:
 - **Increased processor and disk utilization**, leading to better transaction *throughput*
 - E.g., one transaction can be using the CPU while another is reading from or writing to the disk
 - **Reduced average response time** for transactions: short transactions need not wait behind long ones.
- **Concurrency control schemes** – mechanisms to achieve isolation
 - That is, to control the interaction among the concurrent transactions in order to prevent them from destroying the consistency of the database



Schedules

- **Schedule** – a sequences of instructions that specify the chronological order in which instructions of concurrent transactions are executed
 - A schedule for a set of transactions must consist of all instructions of those transactions
 - Must preserve the order in which the instructions appear in each individual transaction.
- A transaction that successfully completes its execution will have a commit instructions as the last statement
 - By default transaction assumed to execute commit instruction as its last step
- A transaction that fails to successfully complete its execution will have an abort instruction as the last statement



Schedule 1

- Let T_1 transfer \$50 from A to B , and T_2 transfer 10% of the balance from A to B .
- A **serial** schedule in which T_1 is followed by T_2 :

T_1	T_2
read (A) $A := A - 50$ write (A) read (B) $B := B + 50$ write (B) commit	read (A) $temp := A * 0.1$ $A := A - temp$ write (A) read (B) $B := B + temp$ write (B) commit



Schedule 2

- A serial schedule where T_2 is followed by T_1

T_1	T_2
read (A) $A := A - 50$ write (A) read (B) $B := B + 50$ write (B) commit	read (A) $temp := A * 0.1$ $A := A - temp$ write (A) read (B) $B := B + temp$ write (B) commit



Schedule 3

- Let T_1 and T_2 be the transactions defined previously. The following schedule is not a serial schedule, but it is *equivalent* to Schedule 1

T_1	T_2
read (A) $A := A - 50$ write (A)	read (A) $temp := A * 0.1$ $A := A - temp$ write (A)
read (B) $B := B + 50$ write (B) commit	read (B) $B := B + temp$ write (B) commit

- In Schedules 1, 2 and 3, the sum $A + B$ is preserved.



- | T_1 | T_2 |
|---|---|
| read (A)
$A := A - 50$ | read (A)
$temp := A * 0.1$
$A := A - temp$
write (A)
read (B) |
| write (A)
read (B)
$B := B + 50$
write (B)
commit | $B := B + temp$
write (B)
commit |



- **Basic Assumption** – Each transaction preserves database consistency.
- Thus, serial execution of a set of transactions preserves database consistency.
- A (possibly concurrent) schedule is serializable if it is equivalent to a serial schedule. Different forms of schedule equivalence give rise to the notions of:
 1. **Conflict serializability**
 2. **View serializability**



Simplified view of transactions

- We ignore operations other than **read** and **write** instructions
- We assume that transactions may perform arbitrary computations on data in local buffers in between reads and writes.
- Our simplified schedules consist of only **read** and **write** instructions.



Conflicting Instructions

- Instructions I_i and I_j of transactions T_i and T_j respectively, **conflict** if and only if there exists some item Q accessed by both I_i and I_j , and at least one of these instructions wrote Q .
 1. $I_i = \text{read}(Q)$, $I_j = \text{read}(Q)$. I_i and I_j don't conflict.
 2. $I_i = \text{read}(Q)$, $I_j = \text{write}(Q)$. They conflict.
 3. $I_i = \text{write}(Q)$, $I_j = \text{read}(Q)$. They conflict
 4. $I_i = \text{write}(Q)$, $I_j = \text{write}(Q)$. They conflict
- Intuitively, a conflict between I_i and I_j forces a (logical) temporal order between them.
- If I_i and I_j are consecutive in a schedule and they do not conflict, their results would remain the same even if they had been interchanged in the schedule.



Conflict Serializability

- If a schedule S can be transformed into a schedule S' by a series of swaps of non-conflicting instructions, we say that S and S' are **conflict equivalent**.
- We say that a schedule S is **conflict serializable** if it is conflict equivalent to a serial schedule



Conflict Serializability (Cont.)

- Schedule 3 can be transformed into Schedule 6, a serial schedule where T_2 follows T_1 , by series of swaps of non-conflicting instructions. Therefore Schedule 3 is conflict serializable.

T_1	T_2
read (A) write (A)	
	read (A) write (A)
read (B) write (B)	
	read (B) write (B)

Schedule 3

T_1	T_2
read (A) write (A) read (B) write (B)	
	read (A) write (A) read (B) write (B)

Schedule 6



Conflict Serializability (Cont.)

- Example of a schedule that is not conflict serializable:

T_3	T_4
read (Q)	write (Q)
write (Q)	

- We are unable to swap instructions in the above schedule to obtain either the serial schedule $\langle T_3, T_4 \rangle$, or the serial schedule $\langle T_4, T_3 \rangle$.



Other Notions of Serializability

- The schedule below produces same outcome as the serial schedule $\langle T_1, T_5 \rangle$, yet is not conflict equivalent or view equivalent to it.

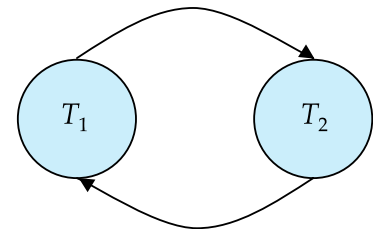
T_1	T_5
read (A) $A := A - 50$ write (A)	read (B) $B := B - 10$ write (B)
read (B) $B := B + 50$ write (B)	
	read (A) $A := A + 10$ write (A)

- Determining such equivalence requires analysis of operations other than read and write.



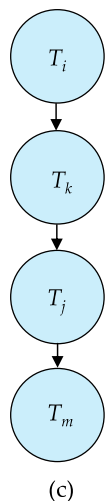
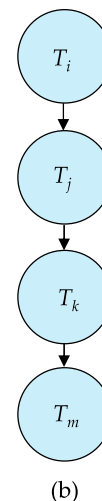
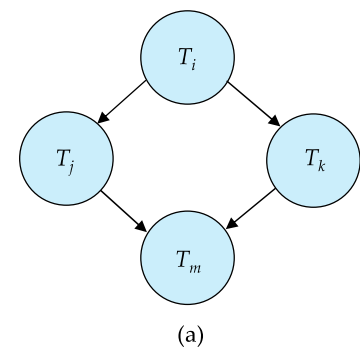
Testing for Serializability

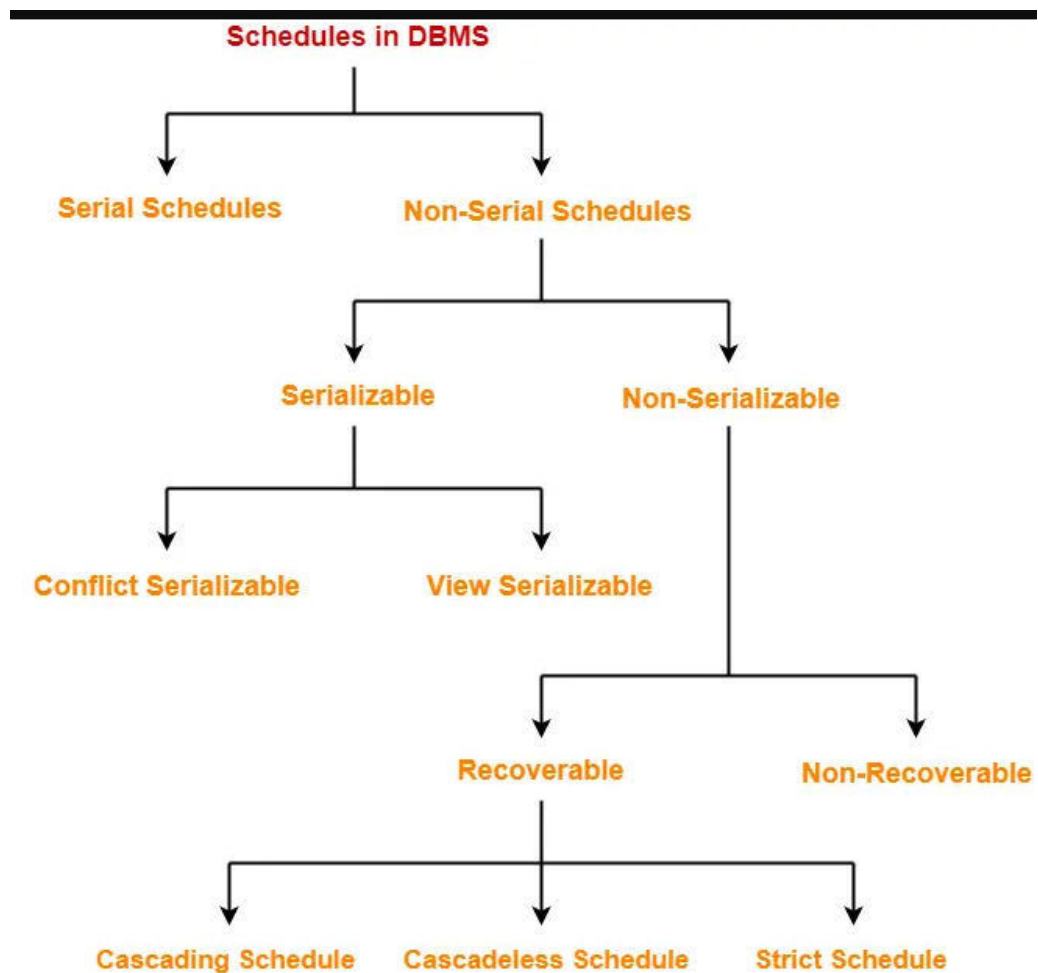
- Consider some schedule of a set of transactions $T_1, T_2, \dots, T_n$
- Precedence graph** — a directed graph where the vertices are the transactions (names).
- We draw an arc from T_i to T_j if the two transaction conflict, and T_i accessed the data item on which the conflict arose earlier.
 - T_i executes $\text{write}(Q)$ before T_j executes $\text{read}(Q)$.
 - T_i executes $\text{read}(Q)$ before T_j executes $\text{write}(Q)$.
 - T_i executes $\text{write}(Q)$ before T_j executes $\text{write}(Q)$.
- We may label the arc by the item that was accessed.
- Example of a precedence graph



Test for Conflict Serializability

- A schedule is conflict serializable if and only if its precedence graph is acyclic.
- Cycle-detection algorithms exist which take order n^2 time, where n is the number of vertices in the graph.
 - (Better algorithms take order $n + e$ where e is the number of edges.)
- If precedence graph is acyclic, the serializability order can be obtained by a **topological sorting** of the graph.
 - This is a linear order consistent with the partial order of the graph.



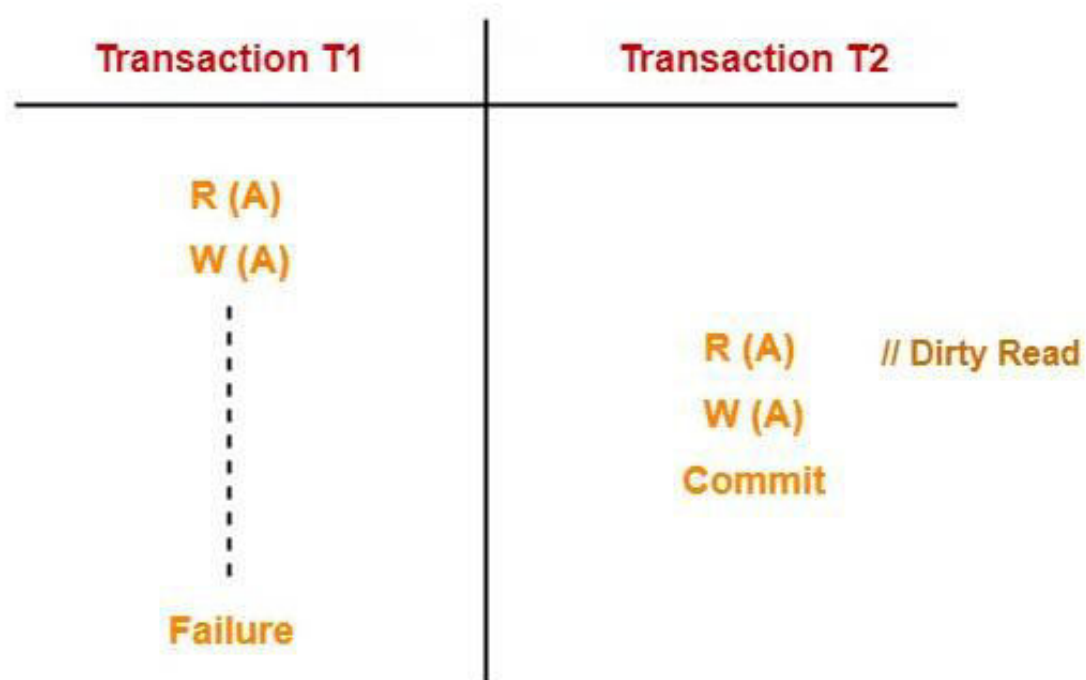


Problems with [Uncontrolled] Concurrency

- Dirty Read
- Unrepeated Read
- Lost Update
- Phantom Event



Dirty Read



Dirty Read

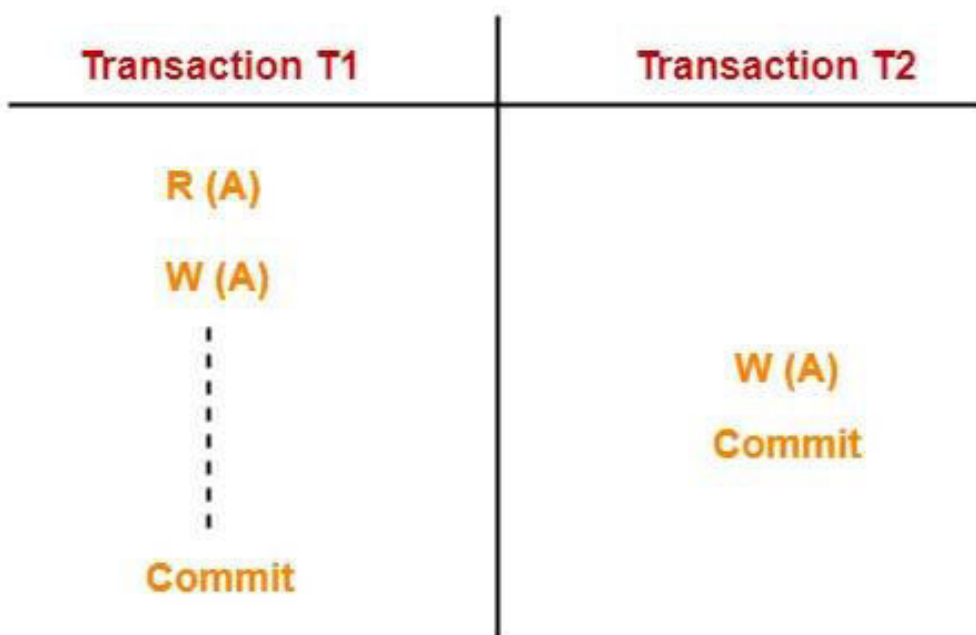
- T2 reads the dirty value of A written by the uncommitted transaction T1.
- T1 fails in later stages and roll backs.
- Thus, the value that T2 read now stands to be incorrect.
- Therefore, database becomes inconsistent



Unrepeatable Read

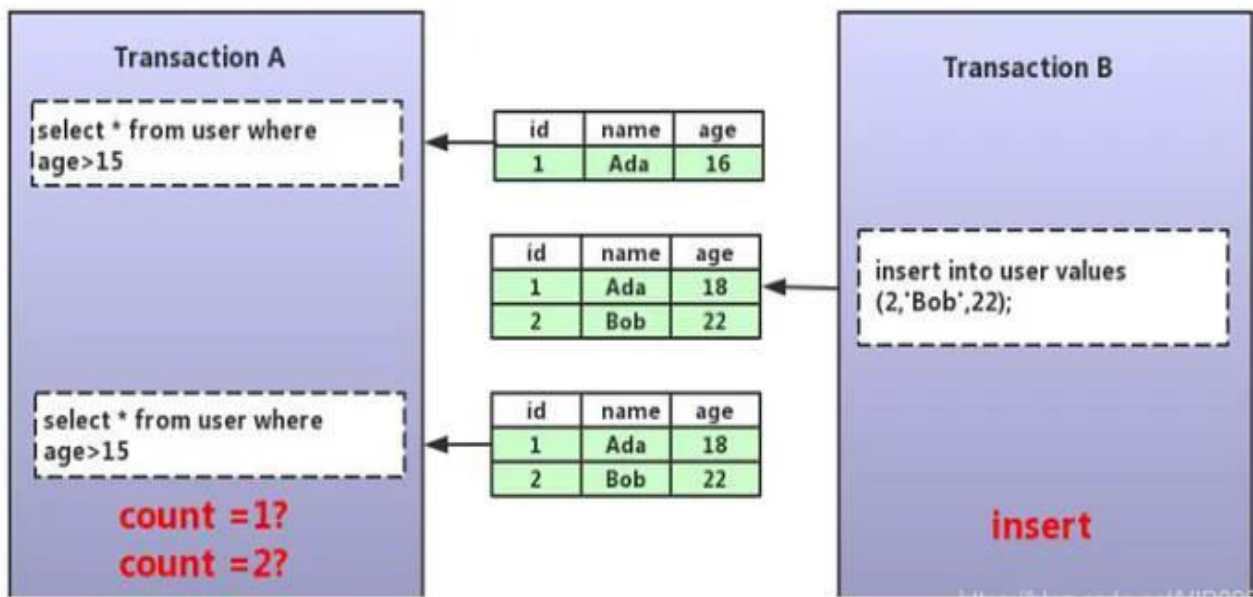


Lost Update Problem





Phantom Event



Transaction B inserts a new row where transaction A reads, then transaction A finds the total count of the result changes to 2



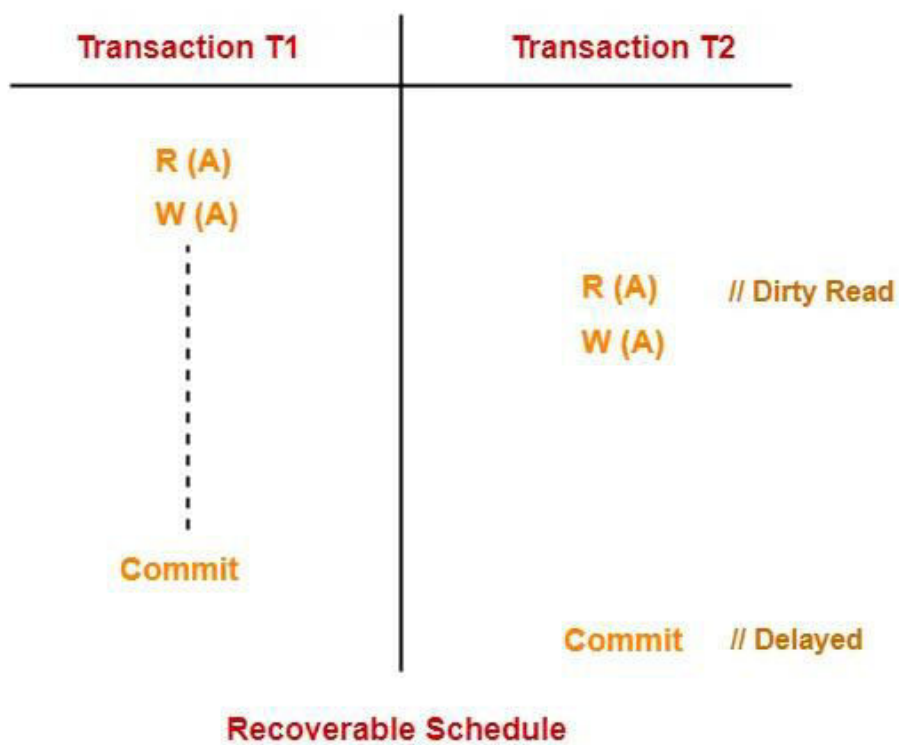
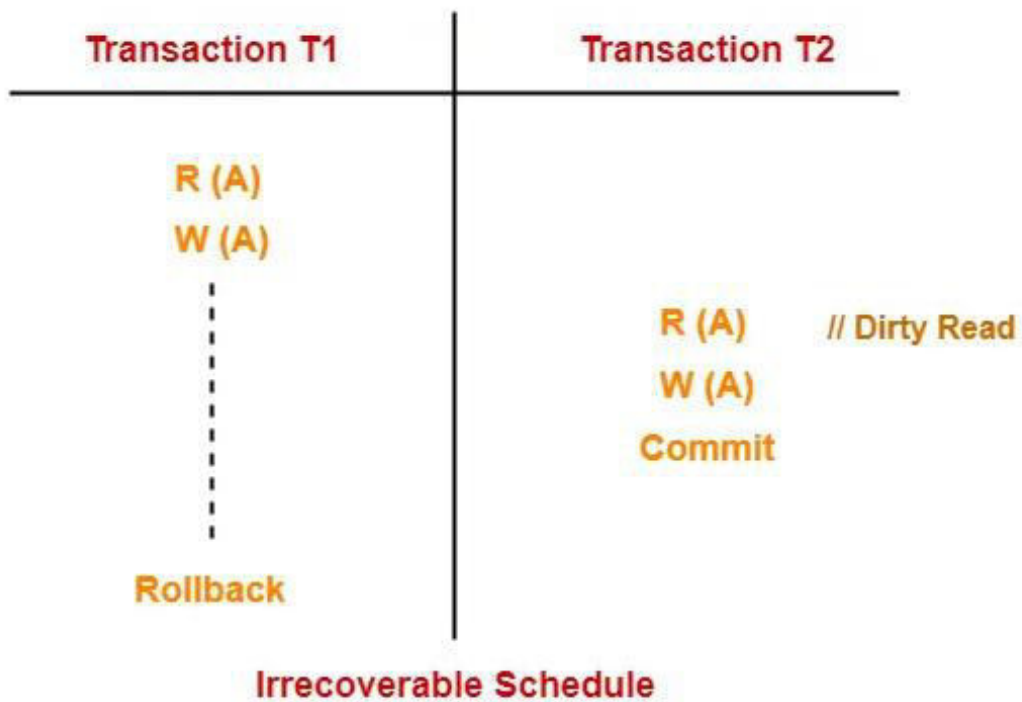
Recoverable Schedules

Need to address the effect of transaction failures on concurrently running transactions.

- **Recoverable schedule** — if a transaction T_j reads a data item previously written by a transaction T_i , then the commit operation of T_i appears before the commit operation of T_j .
- The following schedule (Schedule 11) is not recoverable

T_8	T_9
read (A)	
write (A)	
	read (A)
	commit
read (B)	

- If T_8 should abort, T_9 would have read (and possibly shown to the user) an inconsistent database state. Hence, database must ensure that schedules are recoverable.





Cascading Rollbacks

- **Cascading rollback** – a single transaction failure leads to a series of transaction rollbacks. Consider the following schedule where none of the transactions has yet committed (so the schedule is recoverable)

T_{10}	T_{11}	T_{12}
read (A) read (B) write (A)	read (A) write (A)	read (A)
abort		

If T_{10} fails, T_{11} and T_{12} must also be rolled back.

- Can lead to the undoing of a significant amount of work



Cascading Rollback

T1	T2	T3	T4
R (A) W (A)	R (A) W (A)	R (A) W (A)	R (A) W (A)
Failure			

Cascading Recoverable Schedule

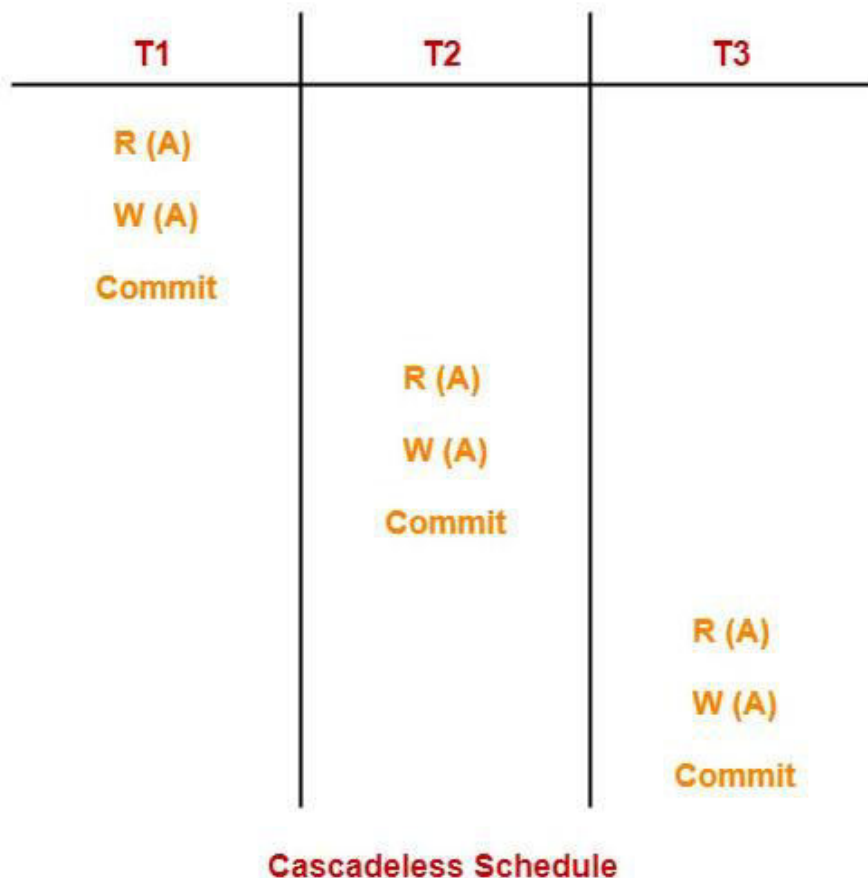


Cascadeless Schedules

- **Cascadeless schedules** — cascading rollbacks cannot occur;
 - For each pair of transactions T_i and T_j such that T_j reads a data item previously written by T_i , the commit operation of T_i appears before the read operation of T_j .
- Every Cascadeless schedule is also recoverable
- It is desirable to restrict the schedules to those that are cascadeless

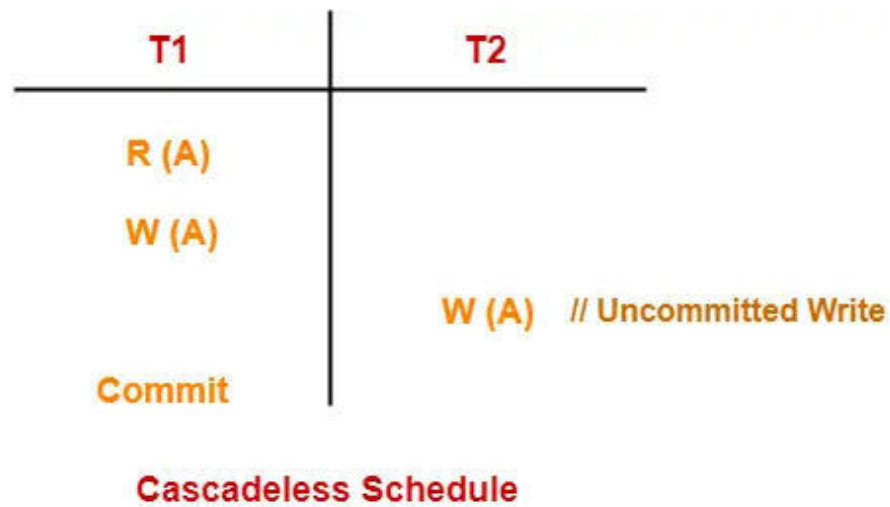


Cascadeless Schedule



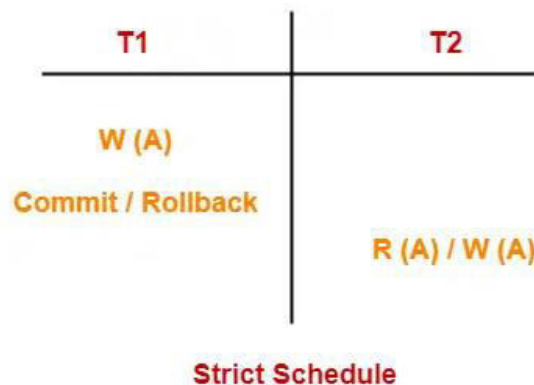


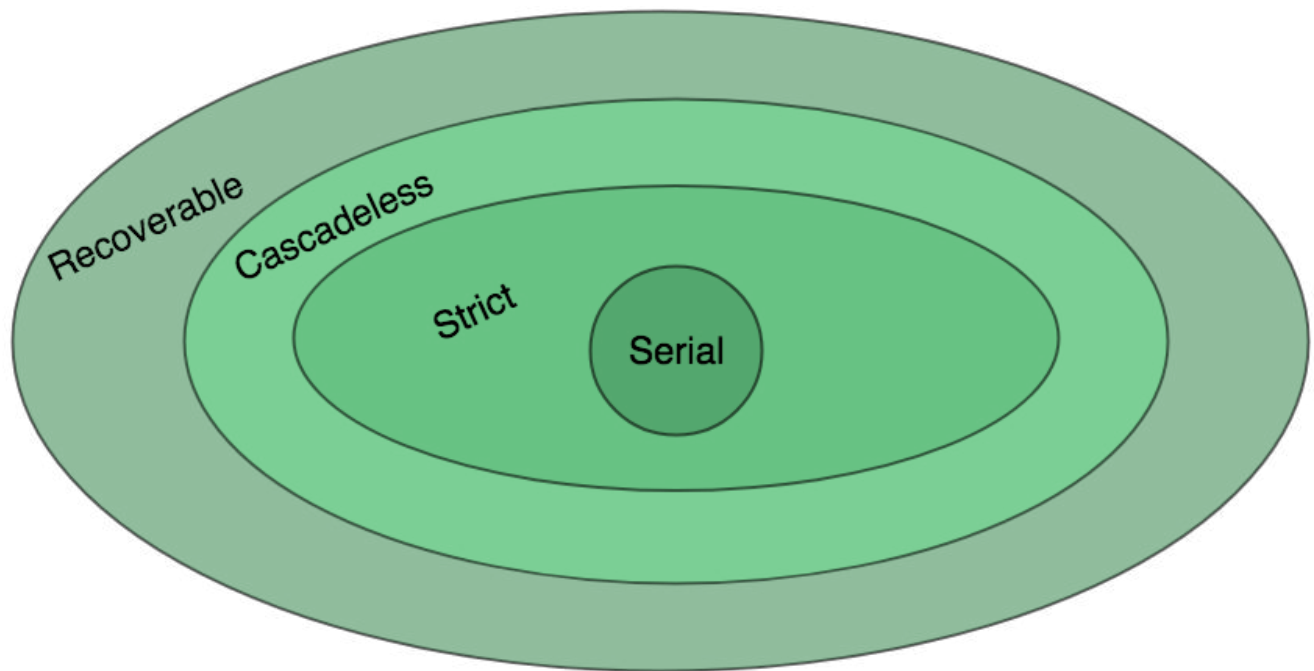
Cascadeless Schedule



Strict Schedule

- If in a schedule, a transaction is neither allowed to read nor write a data item until the last transaction that has written it is committed or aborted, then such a schedule is called as a **Strict Schedule**.
- In other words,
 - Strict schedule allows only committed read and write operations.
 - Clearly, strict schedule implements more restrictions than cascadeless schedule.





	Dirty Read	Lost Update
Recoverable	Yes	Yes
Cascadeless	No	Yes
Strict	No	No
Serial	No	No



Concurrency Control

- A database must provide a mechanism that will ensure that all possible schedules are
 - either conflict or view serializable, and
 - are recoverable and preferably cascadeless
- A policy in which only one transaction can execute at a time generates serial schedules, but provides a poor degree of concurrency
 - Are serial schedules recoverable/cascadeless?
- Testing a schedule for serializability *after* it has executed is a little too late!
- **Goal** – to develop concurrency control protocols that will assure serializability.



Concurrency Control (Cont.)

- Schedules must be conflict or view serializable, and recoverable, for the sake of database consistency, and preferably cascadeless.
- A policy in which only one transaction can execute at a time generates serial schedules, but provides a poor degree of concurrency.
- Concurrency-control schemes tradeoff between the amount of concurrency they allow and the amount of overhead that they incur.
- Some schemes allow only conflict-serializable schedules to be generated, while others allow view-serializable schedules that are not conflict-serializable.



Concurrency Control vs. Serializability Tests

- Concurrency-control protocols allow concurrent schedules, but ensure that the schedules are conflict/view serializable, and are recoverable and cascadeless .
- Concurrency control protocols (generally) do not examine the precedence graph as it is being created
 - Instead a protocol imposes a discipline that avoids non-serializable schedules.
 - We study such protocols in Chapter 16.
- Different concurrency control protocols provide different tradeoffs between the amount of concurrency they allow and the amount of overhead that they incur.
- Tests for serializability help us understand why a concurrency control protocol is correct.



Weak Levels of Consistency

- Some applications are willing to live with weak levels of consistency, allowing schedules that are not serializable
 - E.g., a read-only transaction that wants to get an approximate total balance of all accounts
 - E.g., database statistics computed for query optimization can be approximate (why?)
 - Such transactions need not be serializable with respect to other transactions
- Tradeoff accuracy for performance



Levels of Consistency in SQL-92

- **Serializable** — default
- **Repeatable read** — only committed records to be read.
 - Repeated reads of same record must return same value.
 - However, a transaction may not be serializable – it may find some records inserted by a transaction but not find others.
 - Does online marketplace cart assure repeated read?
- **Read committed** — only committed records can be read.
 - Successive reads of record may return different (but committed) values.
 - Does online marketplace cart assure repeated read?
- **Read uncommitted** — even uncommitted records may be read.



	Dirty Read	Unrepeatable Read	Phantom Event
Serializable	No	No	No
Repeatable read	No	No	Yes
Read committed	No	Yes	Yes
Read uncommitted	Yes	Yes	Yes



Levels of Consistency

- All the isolation levels above additionally disallow **dirty writes**, that is, they disallow writes to a data item that has already been written by another transaction that has not yet committed or aborted.
- Lower degrees of consistency useful for gathering approximate information about the database
- Warning: some database systems do not ensure serializable schedules by default
- E.g., Oracle (and PostgreSQL prior to version 9) by default support a level of consistency called snapshot isolation (not part of the SQL standard)



Transaction Definition in SQL

- In SQL, a transaction begins implicitly.
- A transaction in SQL ends by:
 - **Commit work** commits current transaction and begins a new one.
 - **Rollback work** causes current transaction to abort.
- In almost all database systems, by default, every SQL statement also commits implicitly if it executes successfully
 - Implicit commit can be turned off by a database directive
 - E.g., in JDBC -- `connection.setAutoCommit(false);`
- Isolation level can be set at database level
- Isolation level can be changed at start of transaction
 - E.g. In SQL **set transaction isolation level serializable**
 - E.g. in JDBC -- `connection.setTransactionIsolation(Connection.TRANSACTION_SERIALIZABLE)`



Implementation of Isolation Levels

- Locking
 - Lock on whole database vs lock on items
 - How long to hold lock?
 - Shared vs exclusive locks
- Timestamps
 - Transaction timestamp assigned e.g. when a transaction begins
 - Data items store two timestamps
 - Read timestamp
 - Write timestamp
 - Timestamps are used to detect out of order accesses
- Multiple versions of each data item
 - Allow transactions to read from a “snapshot” of the database



Transactions as SQL Statements

- E.g., Transaction 1:
select *ID, name* **from** *instructor* **where** *salary* > 90000
- E.g., Transaction 2:
insert into *instructor* **values** ('11111', 'James', 'Marketing', 100000)
- Suppose
 - T1 starts, finds tuples *salary* > 90000 using index and locks them
 - And then T2 executes.
 - Do T1 and T2 conflict? Does tuple level locking detect the conflict?
 - Instance of the **phantom phenomenon**
- Also consider T3 below, with Wu's salary = 90000
update *instructor*
set *salary* = *salary* * 1.1
where *name* = 'Wu'
- Key idea: Detect “**predicate**” conflicts, and use some form of “**predicate locking**”



References

- Chapter 17 of Textbook
- <https://www.gatevidyalay.com/concurrency-problems-in-transaction/>
- <https://www.gatevidyalay.com/conflict-serializability-practice-problems/>
- <https://www.gatevidyalay.com/cascading-schedule-cascading-rollback-cascadeless-schedule/>
- <https://www.gatevidyalay.com/recoverable-schedules-irrecoverable-schedules-non-serializable-schedules/>



End of Chapter 17